Electrolyte : Lithium salt in an organic solvent

At the anode oxidation occurs

$Li\ (s) \rightarrow Li^{+}\ (aq) + e^{-}$

At the cathode reduction occurs

$Li^{+} + CoO_2\ (s) + e^{-} \rightarrow Li\ CoO_2\ (s)$

Overall reactions

$Li\ (s) + CoO_2 \rightarrow LiCoO_2\ (s)$

Both electrodes allow Li^{+} ions to move in and out of their structures.

During discharge, the Li^{+} ions produced at the anode move towards cathode through the non – aqueous electrolyte. When a potential greater than the emf produced by the cell is applied across the electrode, the cell reaction is reversed and now the Li^{+} ions move from cathode to anode where they become embedded on the porous graphite electrode. This is known as intercalation.

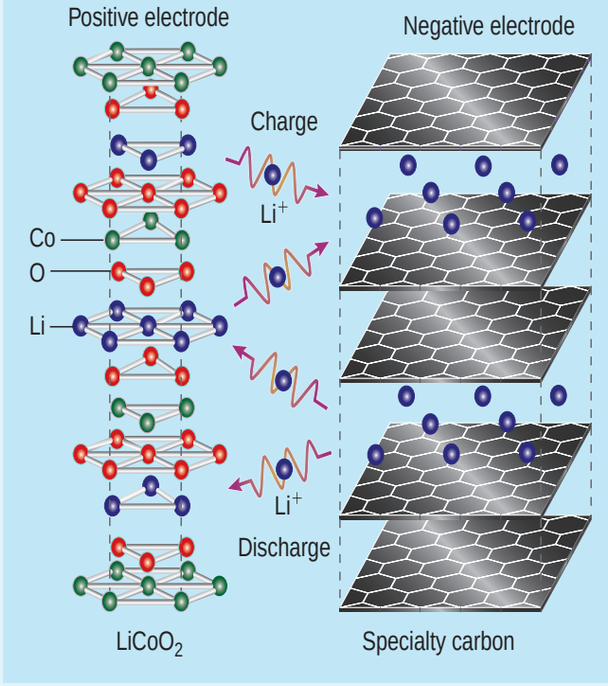


Figure 9.13 Li-ion battery

Uses :

Used in cellular phones, laptops, computers, digital cameras, etc…

Fuel cell

The galvanic cell in which the energy of combustion of fuels is directly converted into electrical energy is called the fuel cell. It requires a continuous supply of reactant to keep functioning. The general representation of a fuel cell is follows

Fuel | Electrode | Electrolyte | Electrode | Oxidant

Let us understand the function of fuel cell by

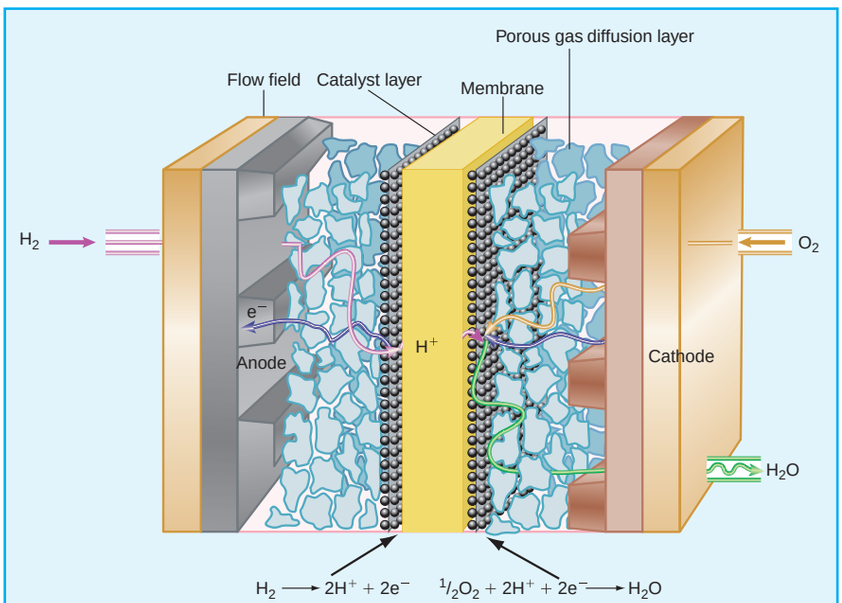


Figure 9.14 H_2O_2 fuel cell

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